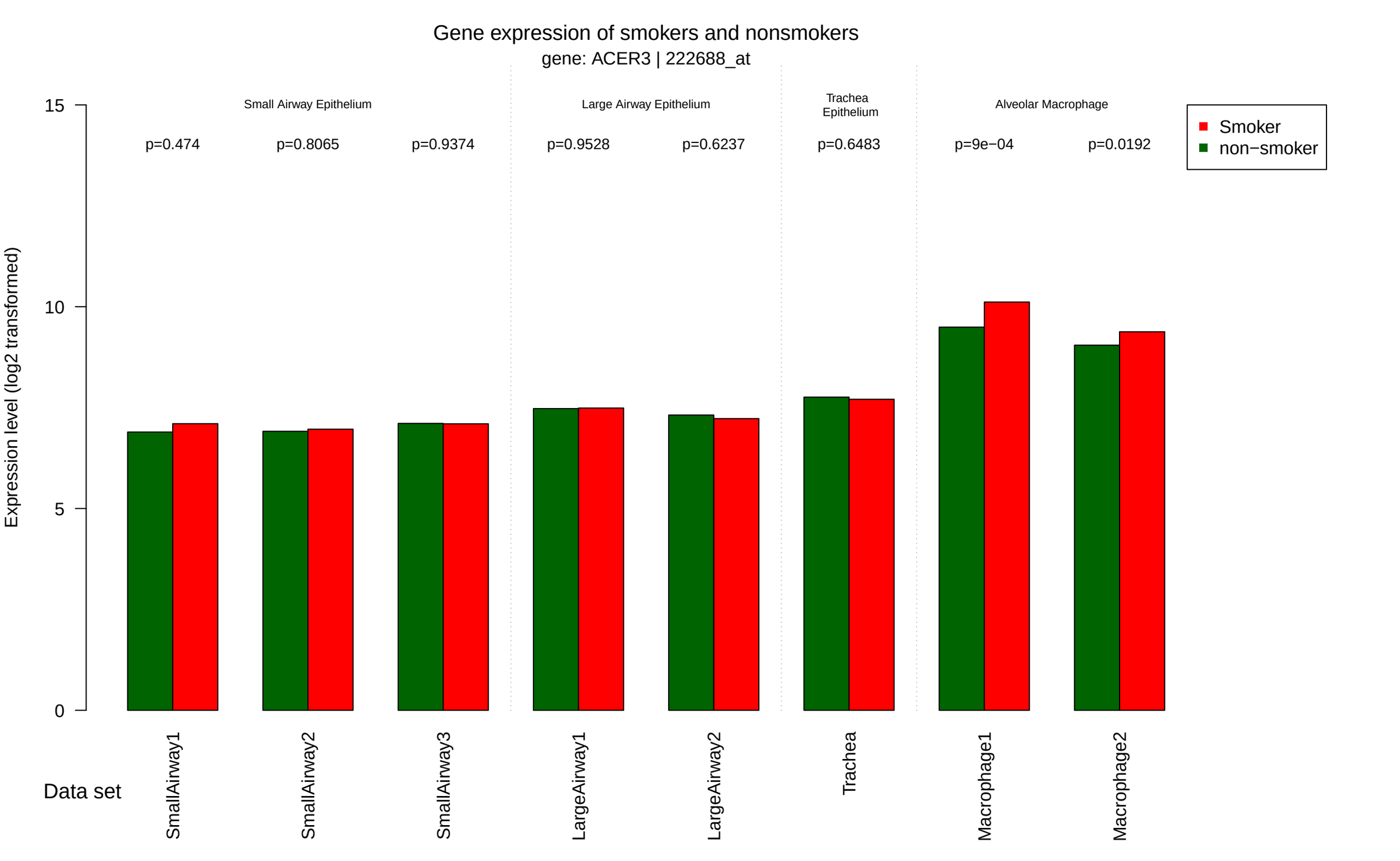
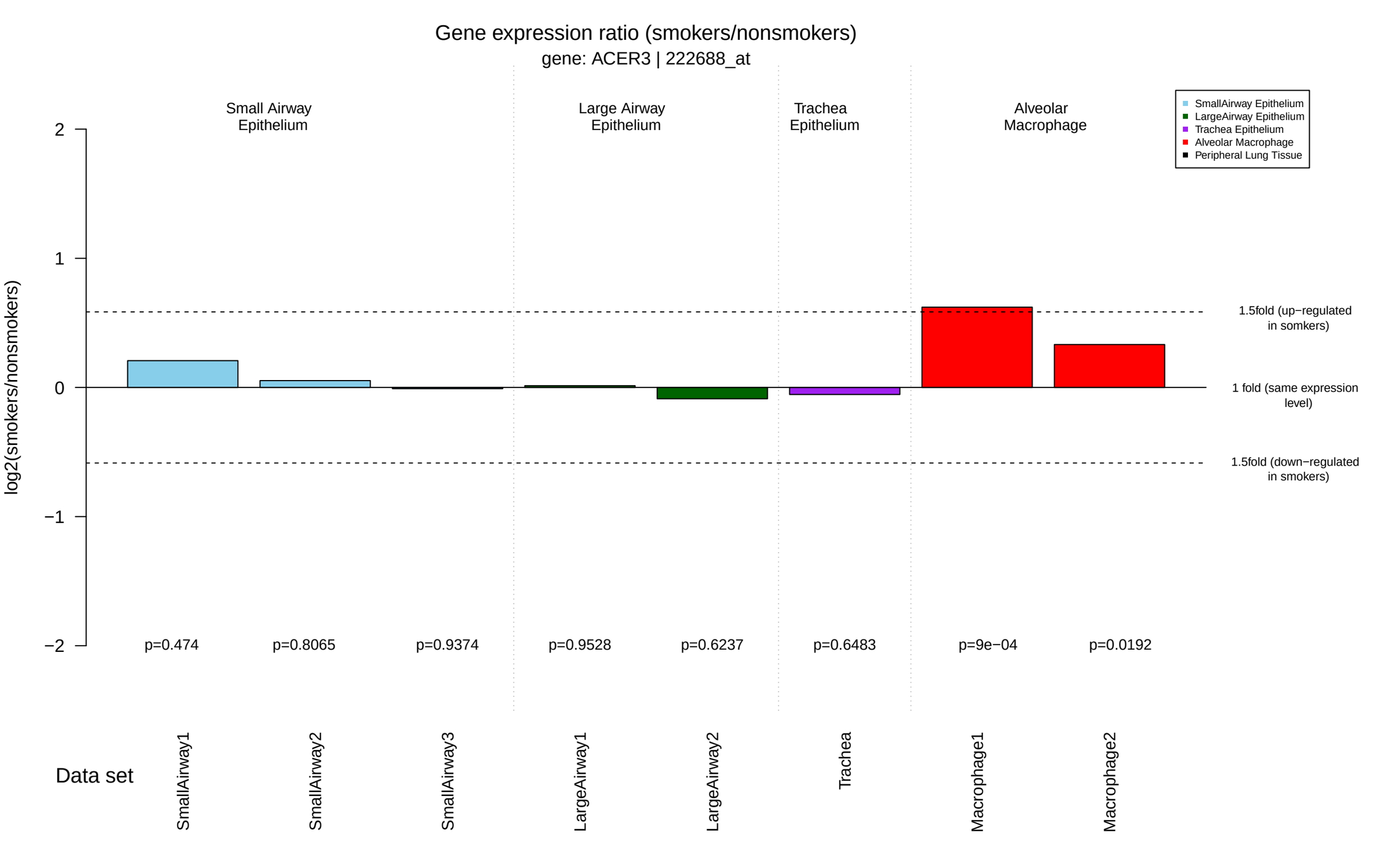
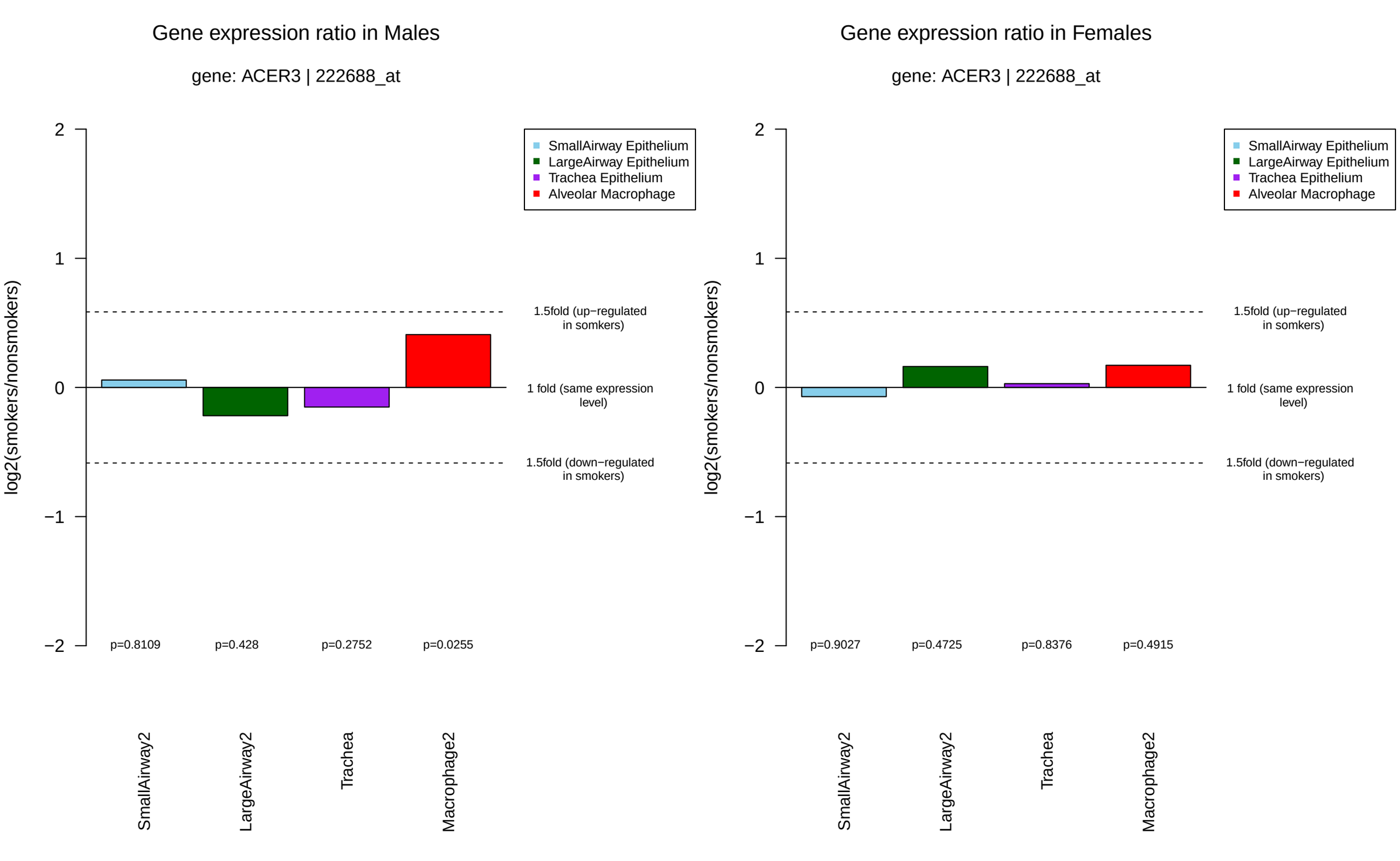
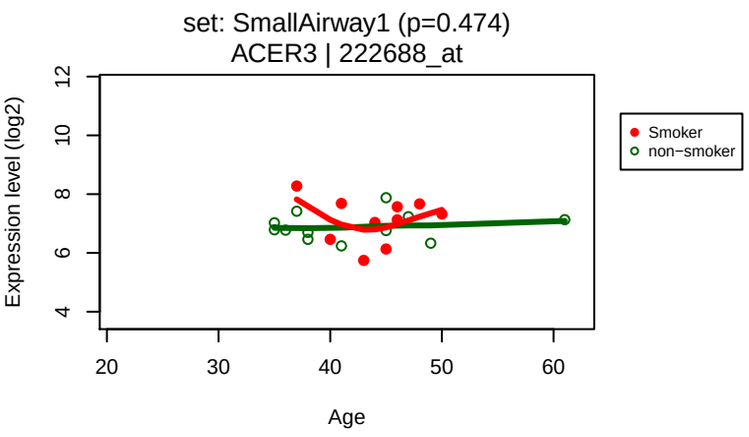
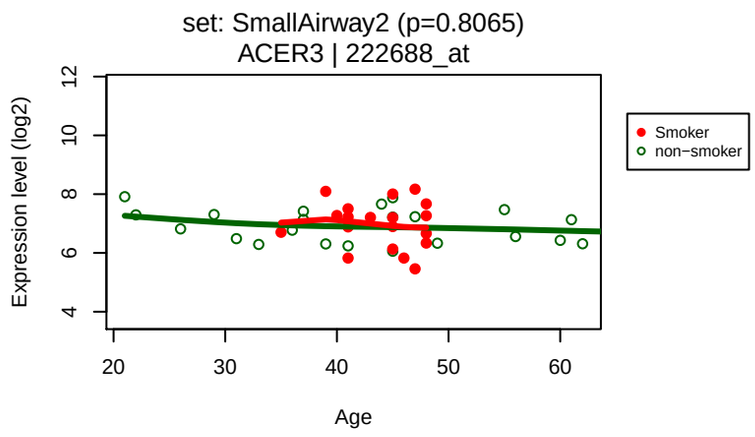
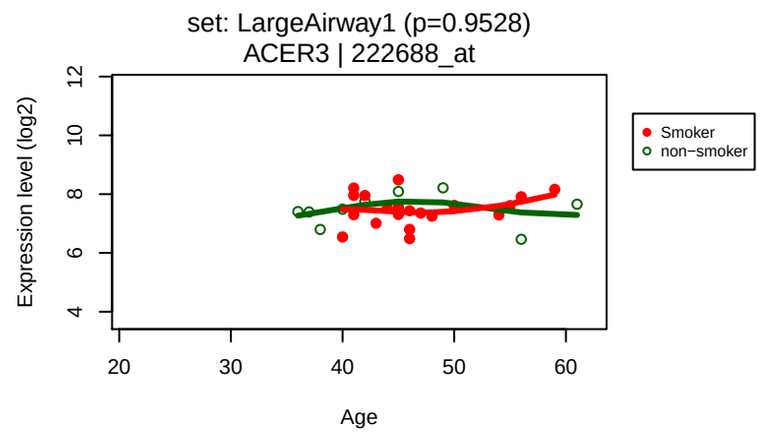

Gene expression of smokers and nonsmokers
gene: ACER3 | 222688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.474
p=0.8065
p=0.9374
p=0.9528
p=0.6237
p=0.6483
p=9e−04
p=0.0192
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

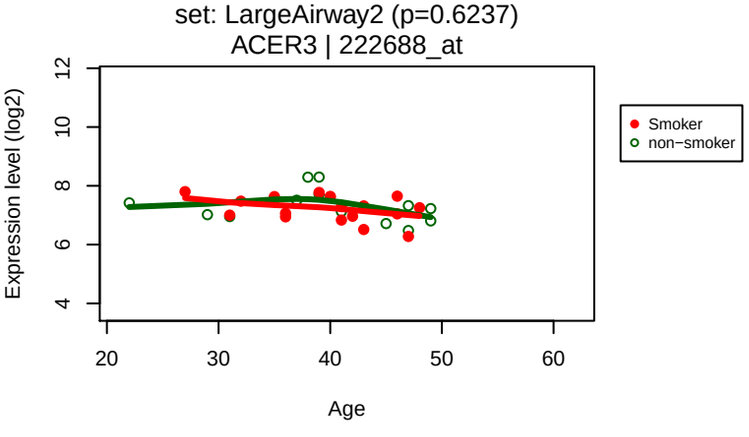

Gene expression ratio (smokers/nonsmokers)
gene: ACER3 | 222688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.474
p=0.8065
p=0.9374
p=0.9528
p=0.6237
p=0.6483
p=9e−04
p=0.0192
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

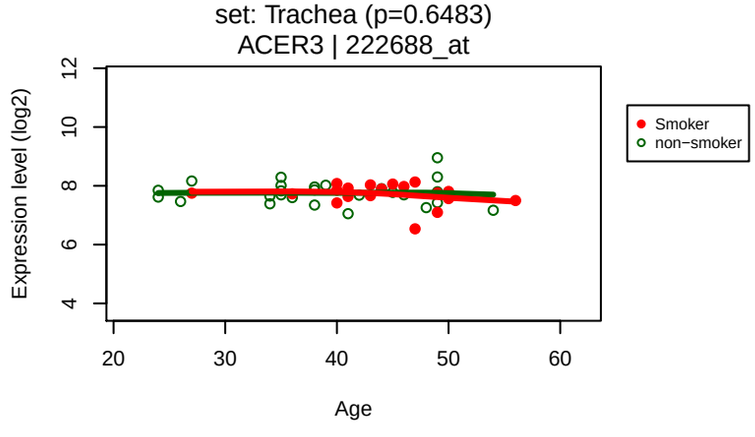

Gene expression ratio in Males
gene: ACER3 | 222688_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8109
p=0.428
p=0.2752
p=0.0255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACER3 | 222688_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9027
p=0.4725
p=0.8376
p=0.4915
SmallAirway2
LargeAirway2
Trachea
Macrophage2

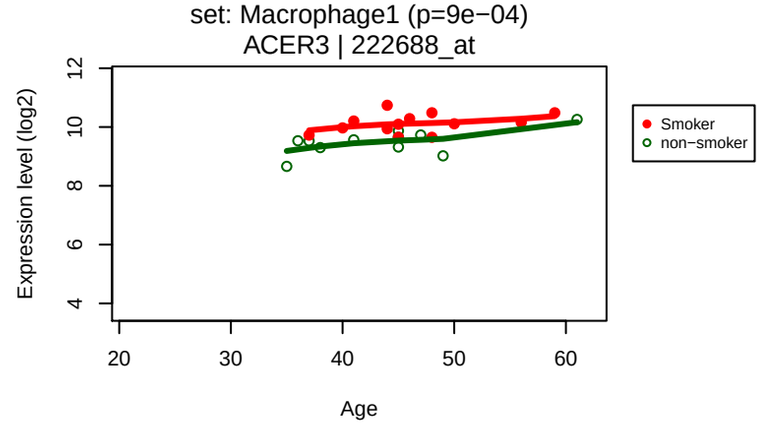

set: SmallAirway1 (p=0.474)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker

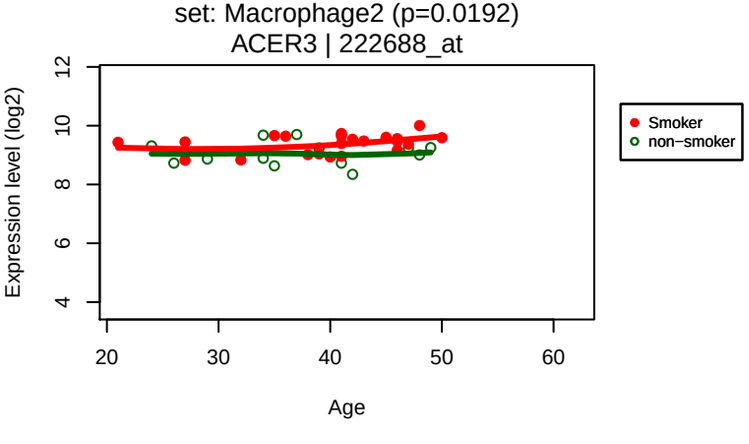

set: SmallAirway2 (p=0.8065)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker

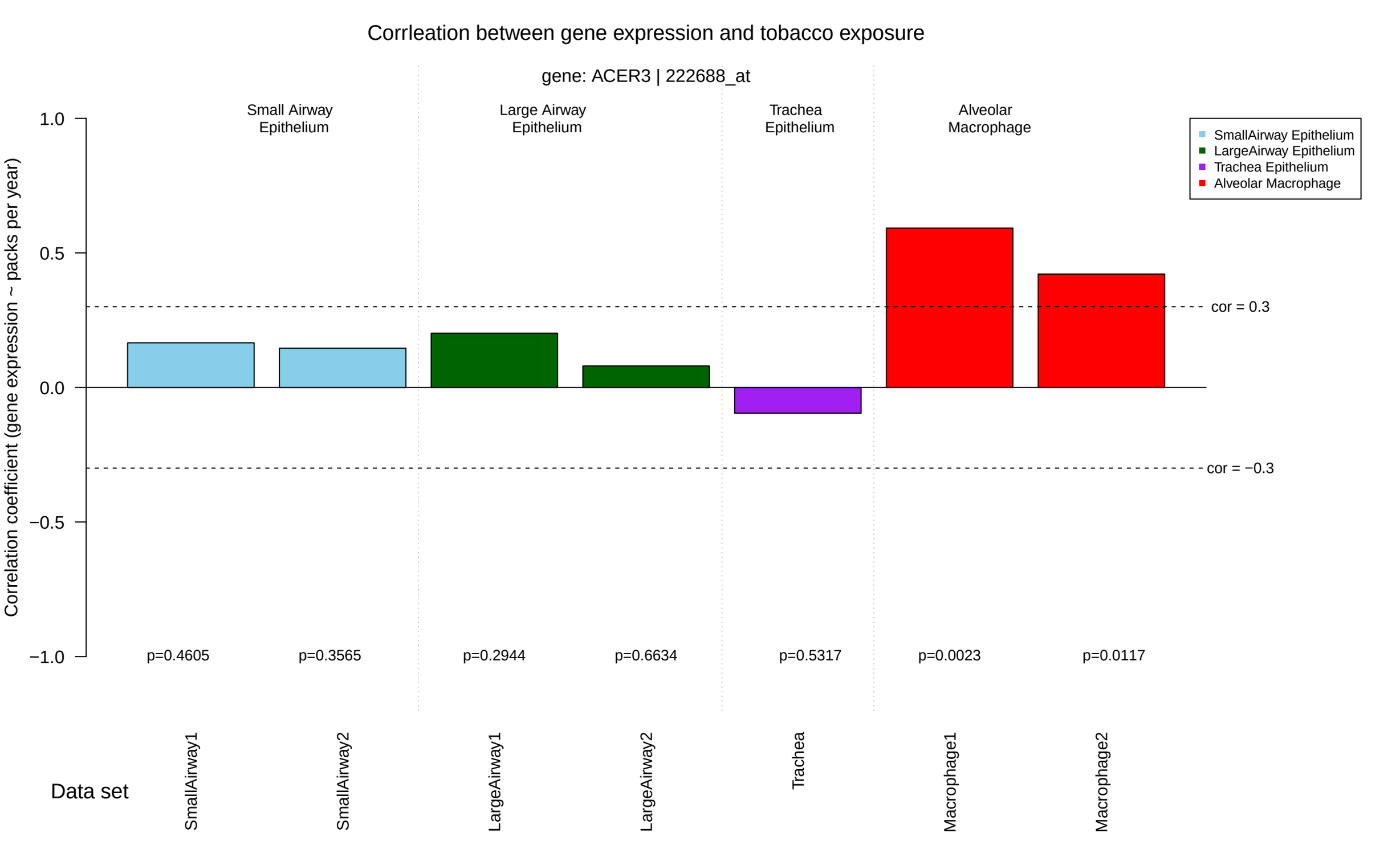

set: LargeAirway1 (p=0.9528)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6237)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6483)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=9e−04)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0192)
ACER3 | 222688_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACER3 | 222688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4605
p=0.3565
p=0.2944
p=0.6634
p=0.5317
p=0.0023
p=0.0117
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2